

Assignment 3: Self-Reflection

This assignment is designed to break the generative AI model GPT-4.1 by making it fail at calculating large multiplications. While the bot initially responds with correct answers, as the numbers grow larger it begins to make mistakes. These failures highlight the way LLM's function and how they differ from calculators or hard coding.

The central reason why the model fails as a precise mathematician is because it does not actually calculate the multiplication, it tries to predict what sequence of tokens are most likely to appear based on the prompt and its training data. For smaller multiplications, the tokens are usually correct because these examples are more common in the training data. As the numbers grow, the mathematical sequence becomes much more uncommon and the model becomes unreliable. Since these models always must have a response, its answers become vastly incorrect, from omitting values to even displaying nonsensical symbols.

Manipulating the model's parameters, such as temperature and top_p, influence the precision of the LLM's responses. Increasing the temperature and top_p makes the responses much more chaotic and random, while lowering it makes the model more consistent. This illustrates that the parameters controlled have no effect over reasoning, only randomness.

This assignment illuminates the limitations of generative AI models such as GPT-4.1 and demonstrates how these LLM's truly function. While they excel at generating language, they lack actual reasoning. The responses are all just statistically calculated tokens that are generated based on observed patterns from training data. AI outputs should never be taken at face value.